



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ORGANIC CHEMISTRY - III				
Type of Course	MAJOR				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Methods of preparation of aldehydes, ketones and carboxylic acids. 2. Nomenclature, isomerism and general physical properties of aldehydes, ketones, carboxylic acids and amines. 3. Acidity of carboxylic acids and basicity of amines.				
Course Summary	To give the students a thorough knowledge about the reactivity, nucleophilic addition and substitution reactions of carbonyl compounds and chemistry of nitrogen containing functional groups and their applications in organic preparations.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyze and compare the reactivity of aldehydes and ketones in nucleophilic addition reactions using various carbon, nitrogen, oxygen and sulphur nucleophiles	An	P	Instructor-created exams / Quiz /Assignment
CO2	Explain the origin of reactivity of carboxylic acids and their derivatives and analyse nucleophilic acyl substitution reactions and hydrolysis of carboxylic acid derivatives	U	C	Class test /Assignment /Quiz
CO3	Demonstrate the oxidation and reduction reactions of alkenes, alkynes, alcohols, aldehydes, ketones and carboxylic acids using various oxidising and reducing agents	Ap	P	Assignment/ Class test

CO4	Identify the preparation methods and important reactions of nitro compounds, amines and sulpha drugs and explain the properties and synthetic transformations of aryl diazonium salts	Ap	P	Assignments /Seminar presentation
CO5	Evaluate reactions involving α -carbons of carbonyl compounds and conjugated addition reactions of α,β -unsaturated carbonyl compounds and apply active methylene compounds in organic preparations	E	C	Class test /Assignment /Quiz
CO6	Conduct qualitative tests to identify specific functional groups, such as aldehydes, ketones, carboxylic acids, phenols, nitro compounds, amines, amides, esters etc. and synthesis of some organic compounds like aspirin, cinnamic acid, iodoform, biodiesel etc	An	P	Lab work/Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Chemistry of Carbonyl Compounds-I		15	32
	1	Origin of reactivity of the carbonyl group. Comparison of reactivity of aldehydes and ketones	2	
	2	Nucleophilic addition reactions of aldehydes and ketones (Mechanism expected) - Carbon nucleophiles (Addition of HCN, Grignard reagents), Nitrogen nucleophiles (NH ₃ , amine, hydroxylamine, hydrazine, semicarbazide and DNP reagent), Oxygen nucleophiles (H ₂ O, alcohols), Sulphur nucleophiles (sodium bisulphite)	3	
	3	Keto-enol tautomerism. Reactions involving α carbons of carbonyl compounds - Aldol condensation, Cannizzaro reaction and Benzoin condensation (mechanism expected), Perkin's reaction, Knoevenagel reaction and Haloform reaction (mechanism not expected)	5	
	4	Conjugate addition reactions of α,β -unsaturated carbonyl compounds – 1,2 and 1,4 – addition reactions	2	

	5	Active Methylene Compounds: Examples – Preparation of ethyl acetoacetate by Claisen condensation (mechanism expected) – Synthetic applications of ethyl acetoacetate	3	
II	Chemistry of Carbonyl Compounds-II		8	18
	6	Origin of reactivity in the carboxylic acid family. Nucleophilic acyl substitution reactions and Mechanism	2	
	7	Comparison of reactivity of Carboxylic Acids and derivatives, Hydrolysis of carboxylic acid derivatives	2	
	8	Fischer esterification (mechanism expected), HVZ reaction, Decarboxylation – Kolbe electrolysis (mechanism expected)	2	
	9	Interconversion of Carboxylic acid derivatives. Introductory idea about β -lactam antibiotics – Structure and action of Penicillin- G	2	
III	Oxidation & Reduction reactions		7	16
	10	Oxidation and reduction of alkenes and alkynes	1	
	11	Oxidation of alcohols with PCC and CrO_3	1	
	12	Oxidation of aldehydes and ketones with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 , CrO_3 ; Oppenauer oxidation. Distinguishing aldehydes and ketones (Tollen's reagent, Fehling's solution)	2	
	13	Reduction of aldehydes and ketones – Catalytic hydrogenation, Wolf-Kishner, Clemmensen, metal hydride (LiAlH_4 and NaBH_4) and MPV reduction	2	
	14	Reduction of carboxylic acids (LiAlH_4 , BH_3)	1	
IV	Nitrogen containing compounds		15	32
	15	Nitro compounds: Preparation and important reactions of nitro compounds, Ketones from nitro compounds – Nef reaction (mechanism not required)	2	
	16	Reduction products of nitrobenzene in acidic, neutral and alkaline media	2	
	17	Amines: Preparation – Gabriel phthalimide synthesis, from reduction of nitriles and isonitriles.	1	
	18	Amines: Properties - Carbylamine reaction, Mannich reaction, Hoffmann's exhaustive methylation, Hofmann-elimination reaction (with mechanism and stereochemistry)	3	
	19	Electrophilic substitution reactions of aniline (Nitration, Bromination and Benzoylation)	2	

	20	Distinction between 1°, 2° and 3° amines with Hinsberg reagent and nitrous acid	1	
	21	Preparation and uses of sulpha drugs – Structural formula of sulphapyridine, sulphadiazine, sulphathiazole and sulphaguanidine	2	
	22	Synthetic transformations of aryl diazonium salts, azo coupling. Preparation of methyl orange	2	
V	Practicals - Reactions of Organic Compounds		30	
	I	Analysis of organic compounds from the following list (also prepare the derivatives). 1. Phenols (phenol, α -naphthol). 2. Nitro compounds (nitrobenzene, o-nitrotoluene). 3. Amines (aniline, N,N-dimethylaniline). 4. Aldehydes and ketones (benzaldehyde, benzophenone). 5. Carboxylic acid (benzoic acid, cinnamic acid, phthalic acid, salicylic acid). 6. Carbohydrates (glucose, sucrose). 7. Amides (benzamide, urea). 8. Esters (ethyl benzoate, methyl salicylate). Analysis of about 7 organic compounds containing the above functional groups.	16	
	II	Organic Preparations (Any two) Synthesis of aspirin, cinnamic acid, iodoform and biodiesel	8	
	III	Open Ended	6	

References:

1. R. T. Morrison, R. N. Boyd, Organic Chemistry, Pearson Education, New Delhi.
2. I. L. Finar, Organic Chemistry, Vol. I, Pearson Education, New Delhi.
3. M. K. Jain, S. C. Sharma, Modern Organic Chemistry, Vishal Publishing Company Co.
4. K. S. Tewari, N. K. Vishnoi, S. N. Mehrotra, A Textbook of Organic Chemistry, Vikas Publishing House.
5. P. Y. Bruice, Essential Organic Chemistry, 3rd Edn., Pearson Education, 2015.
6. John McMurry, Organic Chemistry, 5th Edn., Thomson Asia Pvt. Ltd.
7. C. N. Pillai, Organic Chemistry, Universities Press.
8. Vogel's practical organic chemistry.
9. John McMurry, Eric Simanek, Fundamentals of organic chemistry, 6th Edn., Thomson India Edition.

10. Chemistry for Pharmacy Students: General, Organic and Natural Product Chemistry, S D Sarkar and L Nahar, John Wiley and sons, Ltd.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	1	-	-	-	-	3	1	2	2	2	1	2
CO 2	2	1	-	-	-	-	2	1	1	2	2	2	2
CO 3	2	-		2	-	-	2	2	2	1	2	1	2
CO 4	-	-		2	1	1	2	1	1	2	2	1	2
CO 5	3			2	-	-	2	1	1	2	2	2	2
CO 6	-	-	3	2	-	2	3	2	2	2	2	2	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

Quiz / Assignment/ Quiz/ Discussion / Seminar

Midterm Exam

Programming Assignments (20%)

Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/viva/seminar	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	